

PROJECT DOCUMENTATION

Project Title:

Hotel Maintenance & Repair Tracker

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Introduction

The Smart Hotel Maintenance & Service Management System is a web-based application designed to improve hotel maintenance operations. The system allows hotel staff to report problems, select maintenance workers, track repair progress, and enables administrators to monitor maintenance activities through analytics and reports.

PROJECT DOCUMENTATION

Project Title :

Hotel Maintenance & Repair Tracker

1. Purpose

The purpose of this project is to develop a centralized hotel maintenance management system that helps hotel staff report maintenance issues and allows administrators to manage repair activities efficiently.

The system reduces manual communication between hotel departments and maintenance workers by providing a digital platform for:

- Reporting problems
- Assigning maintenance workers
- Tracking repair status
- Maintaining service history
- Monitoring worker performance

The application improves response time, transparency, and maintenance management efficiency.

2. Problem Statement

In traditional hotel maintenance systems, reporting and tracking repair issues are mostly handled manually through phone calls, messages, or registers.

This creates problems such as:

- Delay in reporting maintenance issues
- Difficulty in assigning the correct worker

- Lack of proper tracking of repair status
- No centralized maintenance history
- Difficulty in analyzing repeated problems

The project solves these problems by providing an automated maintenance tracking system.

3. Goals

The main goals of the system are:

- To create an easy platform for reporting hotel maintenance problems
- To allow users to assign suitable maintenance workers
- To track repair requests from start to completion
- To provide administrators with maintenance analytics
- To maintain previous repair records
- To improve communication between staff and maintenance teams

The system follows the workflow:

Pending → Assigned → In Progress → Completed.

4. Scope

In Scope

The project includes:

User Dashboard

- Login system
- Report maintenance problems
- Enter room number and issue details
- Select priority level
- Upload problem images
- View worker details
- Track request status

- View maintenance history

The report module includes fields such as room number, issue category, description, priority level, and photo upload.

Admin Dashboard

- View total requests
- Monitor pending and completed tasks
- Manage workers
- View analytics
- Track performance
- Monitor high-priority issues

Admin dashboard provides statistics like total requests, pending requests, completed requests, and active workers.

Out of Scope

The following features are not included:

- Online payment system
 - Customer booking management
 - Hotel billing system
 - Employee salary management
 - External vendor management
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5. Users

The system has the following users:

1. Hotel Staff/User

Responsibilities:

- Report maintenance issues
- Select workers
- Track repair progress

- View previous complaints

2. Maintenance Worker

Responsibilities:

- Receive assigned tasks
- Update task progress
- Complete repair work
- Upload repair verification photos

Workers follow task status updates from assigned to completed.

3. Administrator

Responsibilities:

- Manage maintenance activities
 - Monitor workers
 - View reports
 - Analyze maintenance performance
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6. Functional Requirements

Functional requirements describe what the system should do.

User Module

The system should allow users to:

- Login into the system
- Submit maintenance requests
- Enter problem details
- Select issue priority
- Upload images
- View worker availability
- Track request status

Worker Module

The system should:

- Display assigned tasks
- Show room number
- Display issue description
- Allow status updates

Worker task details include room number, issue description, priority level, and assigned date.

Admin Module

The system should:

- Display maintenance statistics
- Monitor worker performance
- Generate reports
- Manage maintenance history
- Identify repeated problems

The system supports analytics like room-wise and department-wise maintenance tracking.

7. Non-Functional Requirements

Performance

- System should respond quickly
- Multiple users should access the system smoothly

Security

- User authentication should be provided
- Data should be protected

Reliability

- System should maintain accurate maintenance records
- Requests should not be lost

Usability

- Interface should be simple and user-friendly

- Users should easily submit complaints

Scalability

- System should support future expansion
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8. Assumptions

The system assumes:

- Users have valid login credentials
 - Maintenance workers have updated availability information
 - Hotel staff provide correct issue details
 - Internet connection is available
 - Administrators regularly monitor requests
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9. Acceptance Criteria

The project will be accepted when:

- ✓ Users can successfully login
- ✓ Users can report maintenance problems
- ✓ Workers can receive assigned tasks
- ✓ Request status updates work correctly
- ✓ Admin can view maintenance analytics
- ✓ Previous maintenance records are stored
- ✓ System handles priority-based issues correctly

The expected outcome is an efficient platform for reporting, assigning, tracking, and analyzing hotel maintenance activities.